Ordinary Generating Functions

Lectures 2–3

1 Product principle

Let f_n be the number of ways to carry out a certain task on the set $[n] = \{1, 2, \ldots, n\}$, and let g_n be the number of ways to carry out another task on $[n]$. Let $F(x)$ and $G(x)$ be the ordinary generating functions of the sequences $\{f_n\}_{n\geq 0}$ and $\{g_n\}_{n\geq 0}$:

$$F(x) = \sum_{n\geq 0} f_n x^n, \qquad G(x) = \sum_{n\geq 0} g_n x^n.$$

Let h_n be the number of ways to split $[n]$ into the two intervals

$$\{1, 2, \ldots, i\} \quad \text{and} \quad \{i+1, i+2, \ldots, n\}, \qquad 0 \leq i \leq n,$$

and then carry out the first task on the first interval and the second task on the second interval. The tasks are understood to depend on the size and order of each interval. If $H(x)$ is the ordinary generating function of $\{h_n\}_{n\geq 0}$, then

$$h_n = \sum_{i=0}^{n} f_i g_{n-i}, \qquad \boxed{H(x) = F(x)G(x)}.$$

Another formulation: widgets and doodles

Assume that a_n is the number of *widgets* (some structures) of size n, and b_n is the number of *doodles* of size n. Write

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots,$$
$$g(x) = b_0 + b_1 x + b_2 x^2 + \cdots.$$

Then $f(x)g(x)$ is the generating function for ordered pairs (u, v), where u is a widget of size i, v is a doodle of size j, and the size of the pair is $i + j$.

Verify:

$$f(x)g(x) = a_0 b_0 + (a_1 b_0 + a_0 b_1)x + (a_2 b_0 + a_1 b_1 + a_0 b_2)x^2 + \cdots.$$

Thus the coefficient of x^n counts precisely the pairs of total size n.

2 Operations on generating functions

We use the notation

$$\langle f_0, f_1, f_2, \ldots \rangle \longleftrightarrow F(x) \qquad \text{when} \qquad F(x) = \sum_{n \geq 0} f_n x^n.$$

1. Scaling

For a constant c,

$$\langle cf_0, cf_1, cf_2, \ldots \rangle \longleftrightarrow cF(x).$$

2. Addition

If $\langle f_0, f_1, f_2, \ldots \rangle \longleftrightarrow F(x)$ and $\langle g_0, g_1, g_2, \ldots \rangle \longleftrightarrow G(x)$, then

$$\langle f_0 + g_0, f_1 + g_1, f_2 + g_2, \ldots \rangle \longleftrightarrow F(x) + G(x).$$

3. Right shifting

Adding k initial zeros corresponds to multiplication by x^k:

$$\langle \underbrace{0, 0, \ldots, 0}_{k \text{ zeros}}, f_0, f_1, f_2, \ldots \rangle \longleftrightarrow x^k F(x).$$

4. Differentiation

$$\langle f_1, 2f_2, 3f_3, \ldots \rangle \longleftrightarrow F'(x).$$

Differentiation has two effects: it multiplies each term by its index, and it shifts the sequence one place to the left. To undo the left shift, apply a right shift:

$$\langle 0, f_1, 2f_2, 3f_3, \ldots \rangle \longleftrightarrow xF'(x).$$

Product

If $\langle a_0, a_1, a_2, \ldots \rangle \longleftrightarrow A(x)$ and $\langle b_0, b_1, b_2, \ldots \rangle \longleftrightarrow B(x)$, then

$$\langle c_0, c_1, c_2, \ldots \rangle \longleftrightarrow A(x)B(x), \qquad c_n = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0.$$

Partial sums

Given any sequence $\langle a_0, a_1, a_2, \ldots \rangle$ with generating function $A(x)$, let

$$s_n = \sum_{i=0}^{n} a_i.$$

The generating function for these partial sums is obtained by multiplying $A(x)$ by $1/(1-x)$:

$$\boxed{\langle s_0, s_1, s_2, \ldots \rangle \longleftrightarrow \frac{A(x)}{1-x}.}$$

3 Extracting coefficients

Taylor's series

Let $F(x)$ be the generating function for the sequence $\langle f_0, f_1, f_2, \ldots \rangle$. Then

$$f_0 = F(0), \qquad \boxed{f_n = \frac{F^{(n)}(0)}{n!}},$$

where $F^{(n)}(0)$ is the nth derivative of $F(x)$ evaluated at $x = 0$. Here the coefficients are taken in characteristic zero, so division by $n!$ is permitted.

4 Linear recurrences and ordinary generating functions

4.1 Fibonacci numbers

The Fibonacci numbers are defined by

$$f_0 = 0, \qquad f_1 = 1, \qquad f_n = f_{n-1} + f_{n-2} \quad (n \geq 2).$$

Their generating function is

$$F(x) = f_0 + f_1 x + f_2 x^2 + f_3 x^3 + \cdots .$$

The first recurrence equations are

$$f_0 = 0, \qquad f_1 = 1, \qquad f_2 = f_1 + f_0, \qquad f_3 = f_2 + f_1, \quad \ldots .$$

Using addition and right shifts, we obtain

$$\begin{array}{rcl} \langle 0, 1, 0, 0, 0, \ldots \rangle & \longleftrightarrow & x, \\ \langle 0, f_0, f_1, f_2, f_3, \ldots \rangle & \longleftrightarrow & xF(x), \\ \langle 0, 0, f_0, f_1, f_2, \ldots \rangle & \longleftrightarrow & x^2 F(x). \\ \hline \langle 0, 1 + f_0, f_0 + f_1, f_1 + f_2, \ldots \rangle & \longleftrightarrow & x + xF(x) + x^2 F(x). \end{array}$$

The sequence in the last line is the Fibonacci sequence. Therefore

$$F(x) = x + xF(x) + x^2 F(x),$$

and hence

$$\boxed{F(x) = \frac{x}{1 - x - x^2}}.$$

The generating function for the Fibonacci numbers is thus a *rational function*: a polynomial divided by a polynomial.

4.2 Getting a closed-form formula

Repeated differentiation of $x/(1-x-x^2)$ looks cumbersome. Instead, use partial fractions to compute the coefficients.

Factor the denominator as

$$1-x-x^2=(1-\alpha_1 x)(1-\alpha_2 x), \qquad \alpha_1=\frac{1+\sqrt{5}}{2}, \quad \alpha_2=\frac{1-\sqrt{5}}{2}.$$

Write

$$\begin{aligned}\frac{x}{1-x-x^2} &= \frac{c_1}{1-\alpha_1 x}+\frac{c_2}{1-\alpha_2 x}\\ &= \frac{c_1(1-\alpha_2 x)+c_2(1-\alpha_1 x)}{1-x-x^2}.\end{aligned}$$

Comparing numerators gives

$$c_1+c_2=0, \qquad -(c_1\alpha_2+c_2\alpha_1)=1.$$

Consequently,

$$c_1=\frac{1}{\alpha_1-\alpha_2}=\frac{1}{\sqrt{5}}, \qquad c_2=-\frac{1}{\alpha_1-\alpha_2}=-\frac{1}{\sqrt{5}}.$$

Using $1/(1-\alpha x)=\sum_{n\ge 0}\alpha^n x^n$, we get

$$F(x)=\frac{1}{\sqrt{5}}\sum_{n\ge 0}\alpha_1^n x^n-\frac{1}{\sqrt{5}}\sum_{n\ge 0}\alpha_2^n x^n.$$

Therefore the formula for the Fibonacci numbers is

$$\boxed{f_n=\frac{\alpha_1^n-\alpha_2^n}{\sqrt{5}}=\frac{1}{\sqrt{5}}\left[\left(\frac{1+\sqrt{5}}{2}\right)^n-\left(\frac{1-\sqrt{5}}{2}\right)^n\right].}$$

5 General linear recurrences

The method for the Fibonacci recurrence can be used for general linear recurrences with constant coefficients, of the form

$$f_n=a_1 f_{n-1}+a_2 f_{n-2}+\cdots+a_d f_{n-d}+g_n, \qquad n\ge d.$$

Homogeneous linear recurrences with constant coefficients give rational generating functions. More generally, the same holds for the displayed inhomogeneous recurrence when the generating function of g_n is rational. Conversely, the coefficients of a rational power series satisfy a homogeneous linear recurrence with constant coefficients for all sufficiently large indices.

Reference in the notes: Martin Aigner, *A Course in Enumeration.*

Transcription notes. Wording, punctuation, and notation have been lightly standardized. On source page 16, the missing factor x in the term $xF(x)$ has been restored. The rationality statement on source page 21 has been qualified to account for the arbitrary forcing term g_n on page 20; arbitrary g_n need not yield a rational generating function. The last source page is blank. No substantive content from the handwritten notes has been omitted.